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Evaluating uncertainties with sample-collection method and source selection in sediment fingerprinting: an example from a Great Lakes tributary

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Abstract

Purpose Identifying suspended-sediment (SS) sources, seasonal variability, and phosphorus (P) transported with SS is critical information for basin managers, although there may be concerns about comparability between flow-integrated SS samples used for sediment fingerprinting and discrete samples used for concentrations and loads in basins where SS is mostly silt + clay and(or) one land-use predominates. Objectives were to determine if (1) sample-collection method and (2) source consideration influenced apportionment of the largest SS source.

Methods Concurrent-replicate, SS samples were collected during 2022 from the East River, Wisconsin using an automated sampler, commonly used for water-quality sampling, and passive samplers, frequently used for SS fingerprinting. Samples were evaluated for differences in physical and chemical characteristics that may affect source apportionment. Considered sources included three upland land-use (cropland, forest, and roads), two in-channel (streambank and streambed sediment), and one that connects uplands to the stream channel (gullies). Source apportionment used established methods in the Sed-SAT tool. Source scenarios included land-use + streambank (4src), 4src + gully, 4src + streambed, and 4src + gully + streambed (6src).

Results There were no statistically significant differences in median grain size, organic carbon, or sediment-bound P as a function of collection method. In-channel sources were the largest proportional SS source, regardless of season, hydrologic condition, collection method, or source scenario. Source verification highlighted which source fingerprints were most accurately defined and implications for SS target apportionment.

Conclusion Varying the source scenarios for sediment fingerprinting indicated that improved management of hydrologic connectivity between upland land use and the stream channel has the potential to mitigate SS loads.

Keywords Sample-collection method · Source confusion · Legacy sources · Source definition

1 Introduction

1.1 Understanding stream-corridor sediment source, punctuated transport, and links with phosphorus is a gap in management of agricultural landscapes

Building on the Great Lakes Water Quality Agreement (International Joint Commission [IJC] 2009), monitoring and research of phosphorus (P) in agricultural Great Lakes basins became more fully recognized and coordinated in the early 2000s. Occurrence and persistence of nuisance and harmful algal blooms (nHABs) were tied to both the

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abundance of dissolved P (DP; a combination of orthophosphate and soluble reactive P), as well as the DP to total P (TP) ratio (Rowland et al. 2021), with toxicity also linked to nitrogen (N) abundance (Jankowiak et al. 2019). Focus basins included the Fox River, Wisconsin, and tributaries (The Cadmus Group 2012), which ultimately flow to Green Bay and Lake Michigan, as well as the Maumee River and tributaries in Indiana, Michigan, and Ohio, which are part of the Western Lake Erie Basin (WLEB; Annex 4 Task Team 2015). The Fox and the Maumee Rivers are similar in that each provide approximately 50% of the annual TP load to downstream Great Lakes, even though water delivery is < 10% of the annual input (Klump et al. 1997; Annex 4 Task Team 2015).

Early mitigation efforts in basins of the Great Lakes prioritized on-field conservation management (U.S. Environmental Protection Agency [USEPA] 2016) that combined improved source nutrient management with reduced delivery of nutrients and sediment from fields to waterways (Sharpley and Jarvie 2012). In the WLEB, total acreage on which structural erosion controls, drainage-water management, and cover crops were implemented was used to model reduced losses of sediment (60–80%) and TP (53–61%) between 2003 and 2012 (U.S. Department of Agriculture - Natural Resource Conservation Service [USDA-NRCS] 2017; Merriman et al. 2019). Water-quality modeling for the Fox River Basin similarly focused on suspended sediment (SS) and TP as pollutants from non-point sources to estimate how reducing agricultural erosion and runoff could mitigate downstream loads (The Cadmus Group 2012; Merriman et al. 2019; The Cadmus Group 2020).

Among lower Fox River tributary basins (e.g., Apple Creek, Plum Creek, and East River; Fig. 1), water-quality modeling (The Cadmus Group 2012) indicated small "natural background" contributions of SS (6–8%) and TP (7–8%), relative to losses from agricultural land (51–72 Mg km^{-2} yr^{-1} SS and 148–178 kg km^{-2} yr^{-1} TP). These findings affirmed prioritization of agricultural conservation management to reduce total maximum daily load targets. The goal was a > 50% reduction by addressing field erosion from storm water and improved management of manure and other nutrient application (The Cadmus Group 2012). Streambank erosion was lumped with non-agricultural, upland sources in these estimates and not considered a major source for two reasons: (1) summer-low streamflow was considered the targeted condition, because it coincides with seasonal nHABs; and (2) there was not sufficient information or model capability to quantify stream-corridor contributions during high-streamflow conditions.

The simplification that sediment and nutrient sources were limited to agricultural land and specific seasons marginalized how timing, process, and transport link on-field

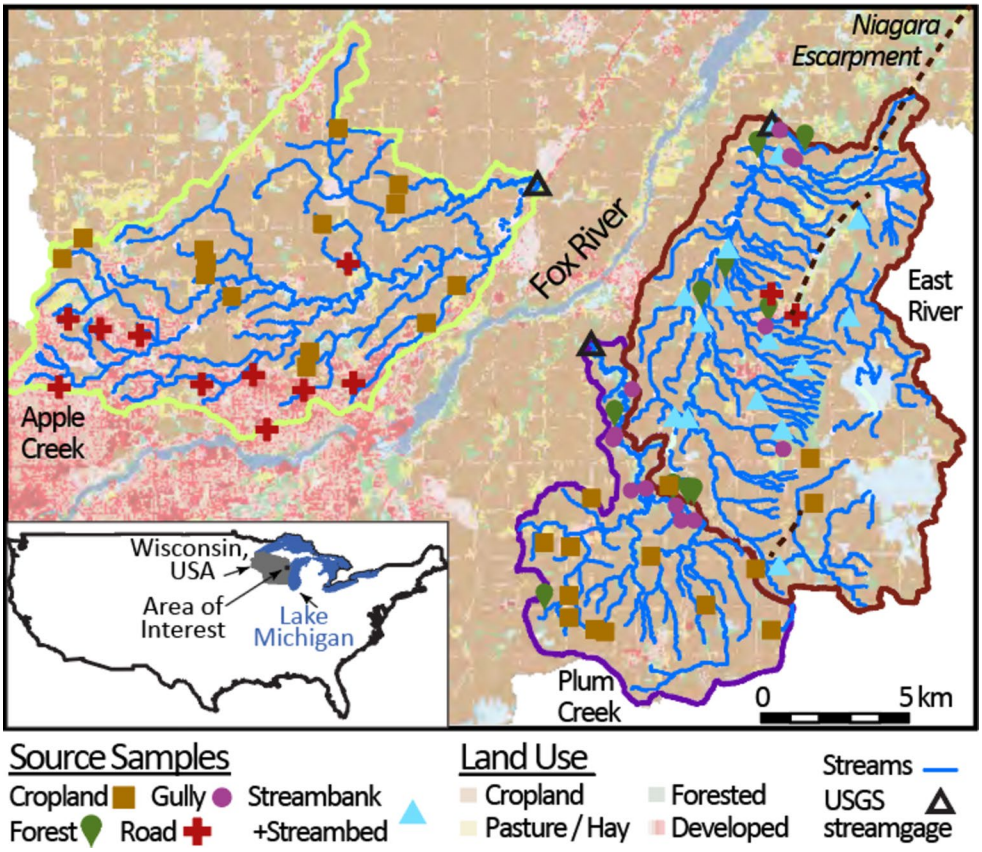

Fig. 1 Area of interest shows location of three Fox River tributary basins, upstream of where the Fox River discharges to Green Bay, Wisconsin and Lake Michigan. Background is 2019 National Land Cover Database (Dewitz and USGS 2024) with source-sample distribution. Approximate location of Niagara Escarpment (Luczaj 2016) is shown for reference. Table 1 summarizes the number of samples for each source type. One streambank and streambed sampling site was co-located with the East River USGS streamgage (Site ID 04085108)

management, the stream-channel network, and environmental response of downstream lakes (Motew et al. 2017; Kreiling et al. 2018). In the East River Basin, annual mean losses from cropland (≥200 Mg km^{-2} yr^{-1} of sediment and ≥600 kg km^{-2} yr^{-1} of TP; Komiskey et al. 2024) exceed in-stream loads (21–89 Mg km^{-2} yr^{-1} SS and 80–320 kg km^{-2} yr^{-1} TP ; U.S. Geological Survey [USGS] 2025). On-field conservation management implementation over several decades has not yet resulted in consistently lower sediment or nutrient loads (Kreiling et al. 2018) nor decreased nHABs occurrence in the Great Lakes (Bosse et al. 2024). Mitigation efforts continue to focus on edge-of- and on-field conditions (Jurjonas et al. 2022), despite more recent watershed plans that acknowledge in-channel sources, including streambanks and accumulated streambed ("bed") sediment, as potential next steps in conservation management (Outgamie County Land Conservation Department 2017; USEPA 2019). Thus, documenting and communicating how in-channel contributions directly influence sediment and nutrient delivery to receiving water bodies can help future management plans.

1.2 Using sediment fingerprinting to inform basin management strategies

It is not unusual for only a small portion of sediment and adsorbed chemicals mobilized from uplands to be delivered to receiving water-bodies (Cashman et al. 2018; Noe et al. 2020). Sediment accumulates along depressions and valley bottoms as well as in the channel (Walling 1983; James 2013; Smith and Wilcock 2015). In agricultural landscapes like the lower Fox River Basin, sediment and P derived from cropland are supplemented by streambank erosion during high streamflow (Fitzpatrick et al. 2019; Blount et al. 2022c). This channel-sourced sediment has a higher likelihood of making it to the watershed outlet (Cashman et al. 2018). Some SS accumulates as bed sediment between runoff events, where it has the potential to be remobilized (Cashman et al. 2018) and to further influence water quality by de-/adsorbing DP (Kreiling et al. 2019). This spiraling of P between sediment and dissolved forms (Newbold et al. 1983; Sharpley et al. 2013) affects the storage and downstream bioavailability of P (Rose and Karwan 2021).

Sediment progresses through the stream system over a series of events (Verhoff et al. 1979; Owens et al. 2005), with size of particles remaining in suspension in balance with streamflow conditions (Wilkes et al. 2019). In Plum Creek, SS source transitioned from upland to remobilized in-channel sources between lower- and higher-order stream reaches (Rose and Karwan 2021). Near the confluence of Plum Creek with the Fox River (USGS Site ID 04084911), SS was > 80% fines (silt + clay; < 63 μm diameter) compared to sources that were 25–50% fines, including both in-channel (streambank & bed) and upland-sourced sediment (Fitzpatrick et al. 2019); median grain diameter (d_{50}) of fine-grained SS was 2–3 μm, with 43–57% clay (<2 μm) (Fitzpatrick et al. 2025). Near the confluence of Apple Creek with the Fox River (USGS Site ID 04085046), fine-grained SS d_{50} was 2–7 μm with 28–45% clay (Blount et al. 2025b). One year of monitoring of the East River showed that 95–100% of suspended material was fines (USGS Site ID 04085108; USGS 2025). Together, these indicate that coarser, upland-sourced sediment may not remain in suspension for downstream transport to higher-order reaches.

Differentiating SS sources using a combination of geochemical indicators (sediment fingerprinting) has been useful for guiding prioritization of basin management strategies (Gellis et al. 2016; Margenot et al. 2023; Ohio Environmental Protection Agency [OEPA] - Division of Surface Water Assessment and Modeling Sect. 2023; USEPA 2024). An advantage of sediment fingerprinting is that it explains the source of a target sediment sample as a proportional combination of different land uses (e.g., cropland, forest, roads & developed land), concentrated flow features (e.g., streambanks, gullies, sub-surface drainage), or geographically defined areas (e.g., tributary basins) in terms that are understandable to stakeholders and resource managers (Owens 2022). Sediment fingerprinting of seasonal- (Crain et al. 2017), monthly- (Sherriff et al. 2018), and event-integrated (Gellis and Noe 2013) SS samples combined with nested (Collins et al. 2010) and paired (Sherriff et al. 2018) basins documented the temporal variability in source contributions. Multi-step sediment-fingerprinting approaches have included re-analysis of targets by re-grouping source samples using land use, soil type, and geologic parent material, which provided consistent ranking of contributing sources (Russell et al. 2001), or combining elemental indicators with weathering indices, which are dependent on geologic parent material (Nosrati et al. 2019). Others compared sediment source for co-located suspended- and streambed-sediment samples (Williamson et al. 2014; Gellis et al. 2017; Cashman et al. 2018) to determine if current sources of SS differ from those in accumulated bed sediment and if this varies longitudinally in the stream network (Koiter et al. 2013; Williamson et al. 2025).

These findings are evidence of landscape connectivity beyond topographic controls and highlight the importance of water-runoff pathways, including erosional features like gullies, and conditions that foster upland versus channel sediment erosion and delivery (Cho et al. 2023). In the case of land-use focused sediment fingerprinting, sediment erosion that occurs outside the boundaries of defined parcels and upslope of the channel network may not match a defined source, resulting in source confusion in the apportionment;

apportionment cannot consider sources for which no sediment fingerprint is provided. Additionally, sediment may accumulate on hillslope features, resulting in seasonal delay or longer-term interruption of sediment delivery to the channel (Belmont et al. 2014), where stream-water is sampled to quantify loads and sources. In contrast, streambank erosion delivers sediment directly to the channel, making it more likely that this sediment will be transported to receiving water bodies and accounted for as a source.

As described by Collins et al. (2020), multiple conditions govern the potential for and utility of sediment-fingerprinting studies as well as certainty of results. First, indicators used for fingerprinting must be relevant to the sources being considered (Buligon et al. 2024) and must be conservative (e.g., moving with sediment, without accumulation or loss) during transport (Pulley et al. 2015; Collins et al. 2017, 2020) which depends on residence time in the stream environment (García-Comendador et al. 2023). Useful geochemical indicators include elements that are environmentally stable and mainly present in sediment-bound forms and not the dissolved phase (Horowitz 1991). Geochemical-indicator conservatism may be influenced by the considered particle-size range (Russell et al. 2001) and how that may combine with organic matter to influence chemical reactivity (Gellis and Noe 2013). A composite of indicator distributions, ranked and weighted using statistical tests, is most effective for discriminating multiple sources (Collins et al. 2010; Gellis and Walling 2011). For example, including geochemical indicators associated with rock-phosphate fertilizers, manure, and vehicles that traverse both fields and roads in the surrounding landscape differentiates agricultural land use (Williamson et al. 2014, 2020), whereas including organic-matter components and elements used in photosynthesis and nitrogen fixation differentiates natural and recovering environments (Brady 1990; Pierzynski et al. 2000; Gellis and Noe 2013; Williamson et al. 2014). Other potential indicator types include fallout radionuclides (Gellis et al. 2019; Rose and Karwan 2021), ratios of stable isotopes (Gorman Sanisaca et al. 2017; Lizaga et al. 2022), and physical properties (e.g., color and size; Boudreault et al. 2019; Pulley and Collins 2021; García-Comendador et al. 2023; Kieta et al. 2023). Combining source datasets from different studies is possible (Williamson et al. 2023), but only if the same indicators are used (Xu et al. 2022).

Second, sediment fingerprinting is constrained by the sources considered, whether based on provenance (geologic parent material; Walling et al. 2008), landscape features (land use and flow paths; Cashman et al. 2018; Xu et al. 2022), or source locations (e.g., waste disposal or burn scars; Kieta et al. 2023). This generally requires an *a priori* identification of relevant sources, analogous to a conceptual model (McMillan et al. 2018) that reflects water and sediment movement in the basin (Pulley et al. 2017; Xu et al. 2022), including surface and sub-surface connectivity of uplands and the stream-channel network (e.g., Russell et al. 2001; Borselli et al. 2008; Gellis and Walling 2011; Porter et al. 2018), and links between geomorphic features and land use (Evrard et al. 2022; Williamson et al. 2024). Sediment fingerprinting of land management generally focuses on small (<300 km^2), headwater basins (Collins et al. 2017). Independent evidence, including field surveys and site-specific monitoring, can help identify potential sources.

Third, sediment apportionment should include an accuracy assessment. This may be evaluated as apportionment of source samples along with target samples (Collins et al. 2010; Gellis and Walling 2011), quantification of the variability or central tendency of indicator abundance (Chalaux-Clergue et al. 2024), or apportionment of synthetic mixtures (either physically or virtually mixed) of source material (Batista et al. 2022), although synthetic mixtures may result in non-conservative indicator abundance relative to source samples (Chalaux-Clergue et al. 2024). Apportionment of individual source samples using the same mixing model that is used to apportion target samples provides information about which sources may be confused (e.g., a streambank source sample being partially or fully apportioned as forest) and underscores those sources which are the most distinct geochemically (e.g., road samples not being misidentified and other source samples not being confused as roads; Williamson et al. 2023). Ultimately, this is confirmation of whether a priori identified sources were differentiable (Pulley et al. 2017).

Fourth, sediment fingerprinting provides a proportional contribution (apportionment) from each considered source for the sampling location. For SS, this is commonly co-located with stream-water monitoring at the basin outlet. These results are most useful when integrated with sediment loads and concentrations that distinguish time periods or locations with the largest amount of mobile sediment (Gellis et al. 2016; Xu et al. 2022). The foundation of this integration is that SS sampled for these different approaches characterizes the same "population" of sediment. Multiple studies have shown the effectiveness of SS collection with passive, time- and flow-integrated samplers to characterize particle-size distribution and mass of sediment (Phillips et al. 2000; Goharrokhi et al. 2019). The collected SS is comparable to what is sampled by autosampler or manual methods, although there is evidence that clay-sized particles disproportionately flow through passive samplers without being collected (McDonald et al. 2010; Panuska et al. 2011).

Finally, sediment fingerprinting may be cost prohibitive (Pulley and Collins 2021). Depending on the approach, a majority of the cost comes from building a fingerprint for each considered source (Williamson et al. 2023). This cost

aspect may affect the type and number of indicators used (e.g., fallout radionuclides, color, elemental abundance; Collins et al. 2020; Pulley and Collins 2021) and the number of source samples collected. Limiting the number of indicators or source samples can increase uncertainty in source differentiation (Buligon et al. 2024).

As acceptance of sediment-fingerprinting studies to identify the largest contributors to sediment loads has become more widespread, so has interest in companion studies in nearby basins where land use differs or geomorphic features are suspected to influence sediment loads. In the WLEB, examples include headwater tributaries in the glacial till and ancient Lake Maumee lakeplain/bed, where differences in soil erodibility led to patterns in cropland distribution, tile-drain density, and concentration of active horse pastures (Williamson et al. 2020, 2025). In the Fox River Basin, agricultural land use transitions to development closer to Green Bay (Fig. 1). This difference in land use combines with post-glacial fluvial erosion and the Niagara Escarpment (Luczaj 2016) to result in differences in valley entrenchment and stream network development (Fitzpatrick et al. 2006; Blount et al. 2023). Previous comparison of source samples from Apple and Plum Creek Basins showed a consistent geochemical signature for roads, cropland, and gullies, reflecting the shared climate, underlying geology, soil associations, and resultant management decisions that influence the sediment fingerprint of these source types (Williamson et al. 2023). This same study indicated the potential for reuse of source samples from these two Fox River tributary basins for sediment fingerprinting of East River suspended sediment.

1.3 Hypothesis and objectives

Integrating on-field and in-stream water-quality monitoring with sediment-fingerprinting studies that include in-channel sources can improve success of subsequent management actions. However, the combination of water-quality modeling (Merriman et al. 2019; The Cadmus Group 2020; Robertson et al. 2023), measured on-field erosion (Komiskey et al. 2024), and fine-grained SS in the Fox River Basin (a relatively even mix of silt and clay) caused concern about refining basin management after sediment fingerprinting indicated that eroding streambanks contributed more SS than cropland (Fitzpatrick et al. 2019; Blount et al. 2022c). The comparability of discrete water-quality samples (used to determine loads and calibrate models) and the SS collected for sediment fingerprinting was identified as a potential methodological misalignment to accurately apportion loads from contributing sources. Another question was the potential for in-channel storage of sediment to cause a temporal disconnect between ongoing upland contributions and what was collected at the basin outlet. Furthermore, continued focus on water-quality management on specific land-use types (e.g., cropland; Jurjonas et al. 2022) overlooked the hydrologic connectivity between uplands and the stream channel after water leaves defined areas.

Ultimately, results from sediment-fingerprinting studies in Fox River tributary basins prompted new questions about why sources known to have active erosion were not identified as large contributors to SS load. Our hypotheses were: (1) that SS collected as part of water-quality monitoring (discrete samples collected using automated-pump samplers) are from the same sediment population as SS collected for sediment fingerprinting (using passive samplers); and (2) that sediment fingerprinting can independently inform management of upland sources, in-channel erosion and accumulation of streambed sediment, and control of SS and sediment-bound P loads. Our objectives were to investigate aspects of SS sediment fingerprinting: (1) to confirm the validity of integrating discrete and flow-integrated methods both in terms of sample characteristics and sediment-fingerprinting results; and (2) to compare different scenarios that combined upland and in-channel sources, provided new information on connectivity of land use to the stream channel, and considered the proportion of SS remobilized from in-channel bed-sediment accumulation. Companion studies independently quantified streambank erosion and bed-sediment storage, including sediment source and associated P, as well as quantifying SS particle-size distribution and P-sorption dynamics in cropland runoff and stream water. These companion studies are not discussed here.

2 Methods

2.1 Study site

Our study site is the upper East River, which is a 115 km^2 basin situated in the Lake Michigan Lacustrine Clay Plain portion of the Southeastern Wisconsin Till Plains ecoregion (Omernik and Griffith 2014). Glacial moraines form much of the drainage divide (Luczaj 2016). The headwaters cut through the Niagara Escarpment (Fig. 1), a semicontinuous ridge of resistant dolostone with generally west to northwest facing steep slopes (Luczaj 2016). Forest commonly marks the escarpment (Howe et al. 2016), and more streambank erosion occurs in downstream reaches (Blount et al. 2025a). Land use in the East River Basin is 77% agricultural, specifically 72% cropland (class 82 in National Land Cover Dataset [NLCD 82] 2019; Dewitz and USGS 2024) and 5% hay [81]. The remainder is 9% development [21–24], 8% forest [41–43], and 6% wetland [90–95]. Manure from local dairy operations where animals are confined is used to fertilize

fields (Merriman et al. 2019), a practice that extends to the Apple and Plum Creek Basins. Conservation management includes a mix of conservation tillage, cover crops, and grassed waterways (Hanrahan et al. 2025).

Individual cropland fields in the East River Basin have 4–8 m of relief (Webber and Williamson 2021). Slopes are generally 2–6%, with areas as steep as 20–30%. The most extensive soil series are Kewaunee (31%), a sandy to silty-clay loam developed on moraines (USDA-NRCS, 2024), Oshkosh (25%) sandy to silt loam developed on glacial lake deposits, and Manawa (10%) silty-clay loam developed in drainage ways. These same soils are also extensive in the Apple and Plum Creek Basins (USDA-NRCS, 2024). These soil series are classified as either soil hydrologic group C or D, indicating that improved drainage can benefit agriculture and the likely presence of subsurface drainage (i.e., tiles).

Streamflow monitoring of the upper East River (USGS Site ID 04085108) began in December 2011, and water-quality sampling began in October 2015 (USGS 2025). During water-year 2022 (WY2022; October 2021-September 2022), the maximum mean daily streamflow was 16 $m^3 s^{-1}$, the second lowest in the period of record and an order of magnitude lower than the historical maximum of 116 $m^3 s^{-1}$ in WY2019. The range of mean daily streamflow during the sampling period of May-November 2022 (Fig. 2a) was 0.005 to 3.15 $m^3 s^{-1}$, with a median daily streamflow of 0.06 $m^3 s^{-1}$ and a maximum instantaneous streamflow of 4.36 $m^3 s^{-1}$. Suspended sediment concentrations ranged from 4 to 933 mg L^{-1}, resulting in estimated daily loads of 0.01–600 Mg (Fig. 2a; Koltun et al. 2006). During the study, monthly loads ranged from 7 to 144 Mg (July and June, respectively; Fig. 2b).

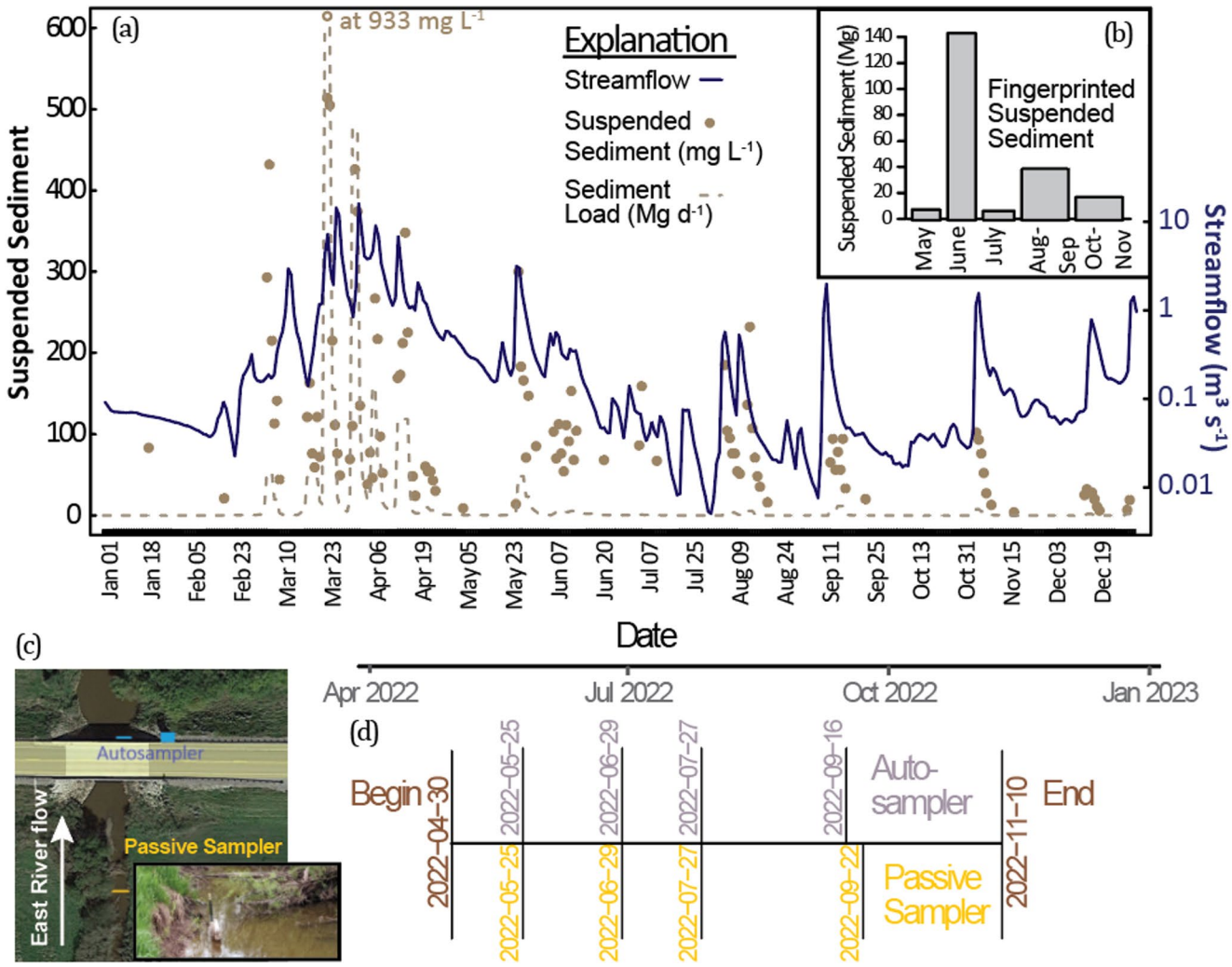

Fig. 2 (**a**) Streamflow, suspended sediment concentrations and estimated daily and (**b**) monthly loads for the East River (USGS Site ID 04085108) during time period when suspended sediment was aggregated for sediment fingerprinting. (**c**) Sediment was aggregated by an AUTOsampler and using a pair of Passive samplers. Location of Passive sampler was approximately 200 ft upstream of the AUTOsampler. (**d**) Dates indicate when samples were collected. Note the log scale for streamflow. Photo credit: Google Earth 2025

Recent analysis of the Fox River Basin (Robertson et al. 2023) showed that SS and TP concentrations and loads decreased between 1989 and 2021, although streamflow, SS, TP, and DP continue to be higher in the lower Fox River, downstream of where a series of agricultural tributaries (e.g., Apple Creek, Plum Creek, and East River; Fig. 1) confluence with the main stem. Field surveys and sediment budgets in Apple (WY2018) and Plum (WY2017) Creeks indicated that erosion estimates of in- and near-channel sources, including streambanks, mass wasting from confined or partly confined valley sides, and gullies, amounted to 100% of measured annual SS loads. Soft, fine-grained, bed-sediment storage amounted to 24% of annual SS loads in Plum Creek (Fitzpatrick et al. 2019) and 30% in Apple Creek (Blount et al. 2022c). Sediment fingerprinting indicated that bed sediment was mostly streambank sourced in these basins, with larger upland contributions in upstream reaches. Near the confluence of Plum Creek with the Fox River, SS was only 50% streambank sourced with the remainder sourced from gullies (23%), eroding roadside ditches (11%), forest (8%), and cropland (6%) (Fitzpatrick et al. 2019). Near the confluence of Apple Creek with the Fox River, SS was 0–94% streambank-sourced, with a larger contribution during higher streamflow, and 0–92% cropland sources, with larger contributions between May and August. Wooded slopes (0–36%) and development (0–60%) contributed the remainder (Blount et al. 2022c). Both studies used established sediment-fingerprinting methods of collection and apportionment (Phillips et al. 2000; Collins et al. 2010; Gellis et al. 2016; Gorman Sanisaca et al. 2017).

2.2 Sediment fingerprinting methods

2.2.1 Suspended-sediment collection

To compare the potential for particle size, sediment chemistry, or source apportionment to differ as a function of collection method, concurrent (bi)monthly time- and flow-integrated SS samples were collected for fingerprinting from May-November 2022 using two different methods (five pairs; Fig. 2): Passive sampler ("Passive") and automated-pump sampler ("AUTO"). All materials used in sample collection and processing were non-metal. To minimize the potential for stream water from later events to affect sediment chemistry, both Passive and AUTO SS samples were collected weekly and dewatered using a streamside centrifuge (Conn et al. 2016). The entire sediment sample from each collection was used; no comparison was made between the mass collected by Passive versus AUTO methods.

Weekly sediment slurries were refrigerated and composited for sediment-fingerprinting analyses for (bi-)monthly periods (Fig. 2). These composites ensured that there was enough material for geochemical analyses. Flow-integrated collection of SS could not begin until ice in the stream channel had thawed. Consequently, sampling began in May after larger spring events, a combination of snow and rain, occurred in April. Multiple small rainfall events were included in the sampled period.

Passive Approximately 30 m upstream of the East River stream gage, a slightly modified version of a Phillips et al. (2000) sampler was placed close to the center of the channel and permanently situated 0.2 m above the streambed (approximately 0.6 times the mean water depth; Fig. 2c). Samplers were constructed from polyvinylchloride (PVC) pipe (10 cm x 1 m long), with 4-mm (inner diameter) inlet and outlet semi-rigid nylon tubing protruding slightly from both cylinder ends. Both ends of the sampler were flat, with no cone. As water passes from the small-diameter tubing into the large-diameter pipe, water velocities slow by a factor of ≈600 enabling SS to settle and accumulate in the pipe. Four samplers were placed in series, parallel to flow, and collected sediment was composited to help ensure that enough sediment was collected for laboratory analyses. The inlet was submerged during low flow to collect a SS sample in all streamflow conditions.

AUTO Approximately 3 m downstream of the bridge at the upper East River gage (Fig. 2c), a second automated-pump sampler was set up alongside that used to collect discrete samples for concentrations and loads (Stuntebeck et al. 2008). The stream is approximately 10-m wide at the gage and the autosampler pumped stream water through a 9.5 mm diameter tube with the inlet placed 3 m from the bank and 0.1 m above the streambed. The sampler was programmed to deliver 1-L of water to a 200-L, plastic drum after a specified cumulative flow volume passed. In practice, the targeted inter-sample volume varied among months to obtain enough material for sediment fingerprinting: between 9000 (in October) and 40,600 (May) m^3. Actual sampling intervals of 85 − 13,280 min (9 days) equated to flow volumes of 578 − 63,644 m^3 (SI1).

2.2.2 Collection of source samples

All source samples were collected using plastic hand shovels to avoid any potential for metal contamination (Gellis et al. 2016). Collection equipment was washed between sites using phosphate-free soap and distilled water, with clean gloves and booties worn at each site to avoid cross-contamination among sites or from external sources. Exposed surfaces were sampled to focus on material susceptible to erosion and that would capture the combined signature of land use (Gellis et al. 2016).

Based on previous comparison of source samples from Apple and Plum Creek Basins (Williamson et al. 2023), a sediment-fingerprinting source-verification analysis was used to identify 49 source samples from Apple and Plum Creek Basins (Blount et al. 2025b; Fitzpatrick et al. 2025) for combination with those from East River Basin (Fig. 1). These previous studies followed the same field methods (Gellis et al. 2016). These three basins are within a combined area of <500 km^2 and East and Plum Creek Basins share a drainage divide (Fig. 1).

Upland sources Three NLCD-derived source types were defined, providing an approach to sediment apportionment that prioritized land use over location of the source sample (Boudreault et al. 2019): cropland, forest, roads as a proxy for development. Ten separate upland-source samples from the East River Basin (three cropland, five forest, and two roads) were collected upstream of the upper East River gage (Fig. 1), between June 28 and August 11, 2022. For each cropland and forest source sample, the top 1–2 cm of soil material was composited along 250-m transects with material collected every 10 m, usually along five parallel lengths. East River Basin cropland samples were collected from monitored fields and included corn silage, wheat, and alfalfa under traditional and conservation tillage and cover crops (Hanrahan et al. 2025). Manure was the common nutrient source in these fields. Animal operations in the basin were confined, so no hay pastures were sampled. East River Basin forest samples were collected from deciduous, wooded streamside slopes, similar to what was observed in Apple Creek (Blount et al. 2022c). Forest samples from Plum Creek were from a mix of upland woodlots, slopes, terraces, and floodplains (Fitzpatrick et al. 2019). Sampled roads were paved but uncurbed and included those among agricultural fields. For each road sample, loose material was composited from the road surface along a 150-m transect, collecting material from both sides and the center at 10-m intervals.

Five gullies were sampled near East River reaches that were also sampled for streambank & bed-sediment (described below). These gullies drained a variety of land-uses and were actively eroding from downcutting or head-ward progression in relatively steeply sloped landforms. A composite of 15 point samples was collected within each gully feature; only eroding and exposed gully surfaces were sampled, not accumulated sediment.

In-channel sources A combination of land cover, stream slope and order, and soil-series distribution was used to select a representative subset of 15 reaches within the East River stream network for a geomorphic assessment of bed-sediment accumulation and streambank erosion using methods described by Fitzpatrick et al. (2019) and Blount et al. (2022b). Soft, potentially mobile bed sediment was sampled along inundated 150-m reaches on August 4, 5, and 11, 2022. The top 1–3 cm of bed sediment was collected from pools, near channel margins, and other slow velocity areas behind bars and large wood; only 14 of 15 reaches had soft, mobile sediment. Eroding streambanks (including terraces and valley sides) were sampled along 14 described reaches using a plastic shovel and composited for each reach; one reach had stable streambanks.

All in-channel samples, including bed sediment and streambank, were from the East River. No in-channel samples were used from Apple or Plum Creeks based on previous comparison of source samples from eight basins that showed that streambank samples could not identify each other using sediment-fingerprinting methods (Williamson et al. 2023).

2.2.3 Lab analyses

Sediment samples, including SS (Pace et al. 2025a, d), bed sediment (Pace et al. 2025c), streambank, and upland sources (Pace et al. 2025b), were shipped to the U.S. Geological Survey Kentucky Sediment Laboratory where a split of the sample was wet sieved to isolate fines (silt + clay; <63 µm diameter), the fraction most likely to be transported by water (Lowrance et al. 2006) and affect biochemical reactions (Owens et al. 2005), including P sorption (Sharpley et al. 1992). A bulk sample was analyzed for particle-size distribution at the U.S. Geological Survey Cascades Volcano Observatory Sediment Laboratory using an X-ray absorption particle-size analyzer (SediGraph III) after treatment with sodium hexametaphosphate and sonic dispersion. A portion of the fines was freeze dried for analysis of > 40 metals using a multi-acid digestion and inductively-coupled-plasma mass spectrometry at the U.S. Geological Survey Analytical Chemistry contract laboratory (Briggs and Meier 2002). Method development included a combination of fluvial and glacial sediment. Another portion of the fines was analyzed for organic carbon (OrgC) at the U.S. Geological Survey Upper Midwest Environmental Sciences Center on a Variomax CN analyzer (Bremner and Tabatabai 1971). To combine source samples from the Apple (Blount et al. 2025b) and Plum (Fitzpatrick et al. 2025) Creek Basins with those from East River Basin, only 38 elements were used as geochemical indicators (SI3). Neither d_{50} nor OrgC were included because they were used to adjust geochemical-indicator concentrations for sediment fingerprinting (described in "Source apportionment"). Together, these geochemical indicators included those most likely to

be present in sediment-bound forms (Meybeck and Helmer 1989; Horowitz 1991) and represent biological processes as well as vehicular, industrial, and agronomic use (White et al. 2024).

A replicate was included for Apple Creek crop sample ACCR06[R]20180424 and evaluated as part of source verification. Particle-size replicates were available for the June and AugSep (August + September composite) Passive samples. These replicates were analyzed independently for source verification and apportionment. There was not enough material for the OctNov (October + November composite) AUTO sample to analyze particle size, so it was apportioned using both the OctNov Passive-sample d_{50} and using the AugSep AUTO-sample d_{50}. Consequently, there is less confidence in apportionment for this sample because d_{50} is used to adjust concentrations of geochemical indicators.

2.2.4 Source scenarios

Apportionment scenarios were considered for each SS target sample, using four different source scenarios (Table 1 and SI2), to evaluate the potential of combining land use (cropland, forest, roads), connectivity features (gullies), and in-channel (streambank and bed sediment) sources. Gullies and bed sediment were not included in all scenarios.

Source sets varied only by the source types included, using the same source-specific samples, with the four apportionment scenarios providing different ways of evaluating sediment source and path. Bed sediment was considered a potential source to discern the proportion of sediment remobilized from in-channel accumulation relative to streambank eroded during each sampling period. Bed-sediment samples are commonly evaluated as sediment-fingerprinting targets (Xu et al. 2022). However, previous studies have shown different source apportionment of contemporaneous and co-located suspended and bed sediment (Williamson et al. 2014, 2020; Gellis et al. 2017), and sediment transported in tiles has been differentiated from cropland-sourced sediment (Russell et al. 2001), indicating distinct geochemical fingerprints as a function of hydrologic environment.

2.2.5 Source apportionment

For each scenario, source apportionment and an accompanying source verification was run for each SS target sample using the Sediment Source Assessment Tool (SedSAT; Gorman Sanisaca et al. 2017). SedSAT encapsulates established methods of concentration adjustment for particle size and organic-matter content and removal of source-sample outliers (Gellis et al. 2014), removal of non-conservative geochemical indicators using a bracket test that compares target concentrations to those from the combined sources (Gellis and Walling 2011), and development of a target-specific forward stepwise linear discriminant function analysis (LDA; step 7 in SedSAT) to unmix a target sample as a function of the considered sources using a composite fingerprint of selected indicators. The resultant mixing model weighted indicators based on the ability of each

Table 1 Source scenarios used for suspended-sediment fingerprinting. Outliers were consistent among source scenarios. Samples from East River are named with land use and those from Apple Creek (AC) and Plum Creek (PL-) are indicated. Of the 38 geochemical indicators provided for the forward stepwise linear discriminant function analysis (LDA), 7–15 were used for individual samples (Fig. 4b). Median and mean apportionment are provided across all months and both sampler types. Streambank is highlighted for each source scenario as the largest proportional contributor. GoF, goodness of fit derived from target-specific source verification (Fig. 5)

Source scenario (src)	Sources included	Source samples provided | used	Samples identified as outliers and removed from provided samples	Number of indicators used in LDA	GoF range	Median | Mean apportionment
4src	Cropland (C), forest (F), roads (R), streambank (S)	65 | 61	Crop-03 [low Ti] ACCR0120180424 [high K] PL-CR02 [high Fe] PL-CR11 [high Th]	7–13 [19 total among all samples]	0.942–0.974	0% | 2% C 13% | 15% F 0% | 0% R **84% | 82% S**
6src	4src + gully (G) + bed (Bd)	92 | 86	Crop-03, ACCR0120180424, PL-CR02, PL-CR11 + Gully-04 [high Zn] + Bed-02 [high Zn & Bi]	10–15 [22]	0.929–0.975	0% | 2% C 16% | 16% G 0% | 11% F 0% | 1% R **69% | 52% S** 6% | 17% Bd
4src + G	4src + gully	78 | 73	Crop-03, ACCR0120180424, PL-CR02, PL-CR11, Gully-04	9–13 [22]	0.929–0.974	0% | 1% C 12% | 13% G 0% | 10% F 0% | 1% R **84% | 75% S**
4src + Bd	4src + bed	79 | 74	Crop-03, ACCR0120180424, PL-CR02, PL-CR11, Bed-02	9–13 [25]	0.932–0.982	0% | 3% C 14% | 20% F 0% | 1% R **81% | 72% S** 0% | 5% Bd

to independently differentiate sources (Collins et al. 2010). For reproducibility, each SS target was analyzed independently using recommended statistical thresholds in SedSAT and a 1000-iteration Monte Carlo analysis (MC1000). This is the same methodology used for fingerprinting of Apple Creek (Blount et al. 2022c) and Plum Creek (Fitzpatrick et al. 2019) sediment as well as previous comparison of source samples from these basins (Williamson et al. 2023). Although there is some evidence that the combined adjustments of both d_{50} and OrgC can introduce uncertainty (Xu et al. 2022), those adjustments were used here to be consistent with the Apple Creek and Plum Creek sediment fingerprinting. Investigating the effect of these adjustments was beyond the scope of work.

Source verification (Collins et al. 2010; Gellis and Walling 2011) was repeated for each source scenario and used the error analysis (step 8 in SedSAT) from individual SS-sample apportionments. Along with each SS target sample, individual source samples are omitted as a source and MC1000 apportioned using the remaining source samples and the target-specific LDA. For each source sample, the MC1000 mean apportionment associated with individual target samples was then averaged to identify the source type with the maximum proportion; effectively, this mean maximum apportionment was used to assign the source sample a source type. In practice, accurately identified source sample were apportioned with > 50% in the proper source category using target-specific LDA.

Thirty-eight tracers were used to differentiate source types (unadjusted concentrations in SI3). For each source apportionment, allocations of < 0.5% were considered zero. Because an apportionment only considers the sources provided, this multi-step approach of each target being apportioned using multiple source-scenarios (Table 1) was designed to provide insight into source confusion among land-use and landscape features as well as how sediment travels from the uplands to the basin outlet (Evrard et al. 2022). Mixing model error was quantified for verification of individual source samples; this aggregates the difference in tracer concentrations in verification target and source groups as a function of apportionment and incorporates the relative weights of each indicator (Collins et al. 2010; Gellis et al. 2014; Eq. 7 in Gorman Sanisaca et al. 2017). Goodness of fit (GoF) was calculated as

$$1 - (\sqrt{mixing\ model\ error} \div number\ of\ indicators) \quad (1)$$

2.2.6 Statistical methods

Source and target samples were evaluated for differences in d_{50} and OrgC because they combine to influence surface area and reactivity and indicate potential erodibility, mobilization by water from source areas, transport as SS, and accumulation or later remobilization of bed sediment (Gellis and Walling 2011). Sediment-bound P (sed-P; also known as particulate P) is important for mitigating nHABs, and differences in sed-P reflect potential (de)sorption as particles move through the basin. Each of these could be influenced by the collection method.

Source verification for sediment fingerprinting used mean maximum apportionment, with examination of source confusion using the mean MC1000 apportionment of individual source samples for each SS target. Source verification provided information on source confusion (complete or partial apportionment as the wrong source type) both as a sample being misidentified when the mean apportionment was averaged for all SS targets and how partial confusion varied as a function of the source scenario; this was similar to a confusion matrix. Source apportionment of SS targets were evaluated as means for each source scenario and as the range of apportionments among MC1000 iterations. Source apportionments were then compared for individual SS targets both between sampling methods and among the different source scenarios.

Statistical analyses were done in R 4.3 (R Core Team 2025). SedSAT is run in an encapsulated version of R 3.3.0 along with contemporaneous libraries (Gorman Sanisaca et al. 2017). Tukey box plots and pairwise Wilcoxon rank-sum tests (pairwise.wilcox.test with BH correction; Benjamini and Hochberg 1995) were used to compare d_{50}, OrgC, and sed-P as well as apportionment among sources and verification of source types. The scatterplot matrix used the car package (Fox et al. 2007). Statements of significance indicate a p-value ≤ 0.05. Letters included on graphs indicate significantly different groups with *a* differentiated from *b* but *ab* not differentiated from either.

3 Results

3.1 Median grain size, total organic carbon, and sediment-bound phosphorus of samples

There were no statistically significant differences in d_{50}, OrgC, or sed-P between Passive and AUTO SS samples (Fig. 3). May and June samples, which were collected during higher streamflow (0.04–3.17 $m^3\ s^{-1}$), were generally coarser, with lower sed-P and OrgC relative to those collected in later months with lower streamflow (0.005–1.98 $m^3\ s^{-1}$).

Upland and gully source samples from East River Basin, each of which were near the Niagara Escarpment (Fig. 1), were generally coarser than those from Apple Creek and

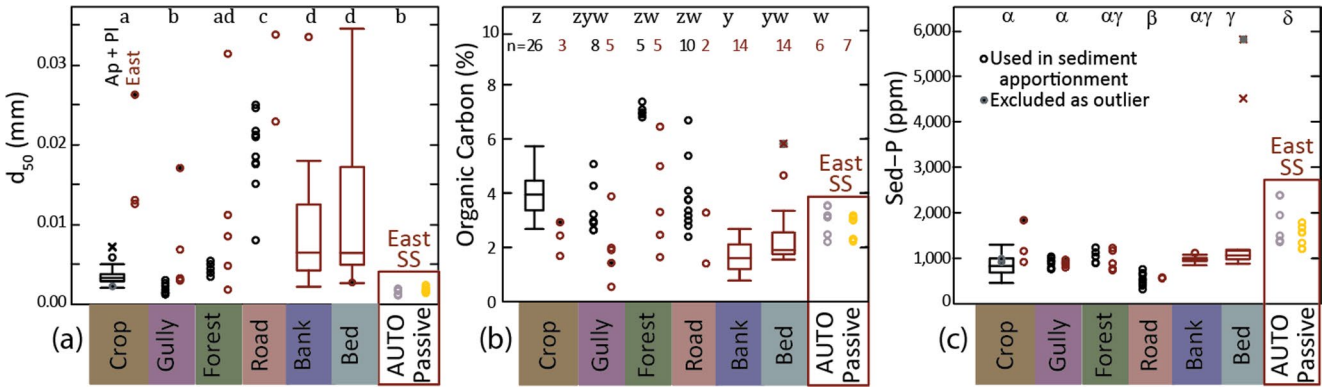

Fig. 3 Median grain diameter (d_{50}), organic carbon concentration (OrgC), and total sediment-bound phosphorus concentration (sed-P) for source (crop, gully, forest, road, streambank [bank], and streambed [bed] and target suspended-sediment [SS] samples). Samples include those from the East River Basin as well as source samples combined from Apple Creek (Ap) and Plum Creek (Pl) Basins, that are proximal Fox River tributaries. There was no statistically significant difference in d_{50}, OrgC, or sed-P as a function of how suspended-sediment was collected. Four cropland, one gully, and one streambed samples were excluded as outliers for sediment fingerprinting of SS. Letters indicate significant differences in sediment characteristics. n, number of samples; AUTO, collected by automated-pump sampler; Passive, collected by in-situ passive sampler

Plum Creek Basins, which were collected from the glacial-sediment derived soils that are also present in the East River Basin (Fig. 3a). This difference was statistically significant for both cropland and gully. The coarsest of each was eliminated as an outlier during sediment fingerprinting (Table 1) due to unadjusted indicator concentrations that were more than three standard deviations from the mean for that source type (step 2 in SedSAT; gray symbols in Fig. 4a). The East River cropland outlier, Crop-03, had the highest cropland sed-P as did the bed-sediment outlier, Bed-02; Bed-02 also had the smallest d_{50} and was later re-assessed as a field swale that was inconsistently connected to the stream network. Except for forest, d_{50} significantly differed among upland sources and relative to in-channel sources. Bed sediment included larger particles than streambank although median d_{50} was similar between the two. Suspended sediment was finer than all sources except gully (Fig. 3a).

East River Basin upland and gully OrgC source-sample concentrations were significantly lower relative to Apple Creek and Plum Creek Basin samples, with the exception of road (Fig. 3b). Combined upland sources had similar OrgC. There were no significant differences between East River streambank and bed sediment. Suspended sediment OrgC was significantly lower than cropland and higher than streambank.

Except for road, there were no significant differences in sed-P among the upland sources (Fig. 3c). Bed-sediment sed-P was significantly higher than most upland sources but overlapped with forest and streambank. Suspended sediment sed-P was significantly enriched relative to all sources. This 2–3 factor difference in sed-P was within what might be expected as a function of the finer grain size (Kleinman et al. 2011).

3.2 Geochemical indicators for sediment fingerprinting

Five geochemical indicators (Ca, Co, Ga, Mg, V) were used in the LDA for all source scenarios, and Sc was used when bed was included as a source (Fig. 4a). Among source scenarios, 31 geochemical indicators were used for apportionment of one or more SS samples (Fig. 4b and SI3). Seven (Ce, Fe, K, P, S, W, Zn) were not identified by SedSAT as being indicative of SS source. Either d_{50} or OrgC significantly correlated with geochemical-indicator concentration for at least one source type for all but Bi, resulting in source-specific adjustment of geochemical-indicator concentrations as part of sediment fingerprinting (Fig. 4c and SI3). Bed and gully were most commonly adjusted. For some geochemical indicators, both d_{50} and OrgC adjustments were done for gully (Pb, U, Zn), road (Mn), or bed (Al, Cr, Ga, S). Phosphorus concentrations were adjusted for bed-sediment d_{50} and streambank OrgC. The only other significant correlation for streambank was K with d_{50}. Regardless of collection method, P was identified as non-conservative using the bracket test for all samples after June. Similarly, Mn was removed in the bracket test for all but the June Passive SS sample with the 4src + G scenario.

3.3 Source verification for sediment fingerprinting and source confusion

There was no significant difference in GoF between Passive and AUTO samples (p-value ≥ 0.2949) or among source scenarios (p-value ≥ 0.6800) (Fig. 5). Source verification of ACCR06[R]20180424 replicates were consistently identified as 100% cropland, though individual geochemical

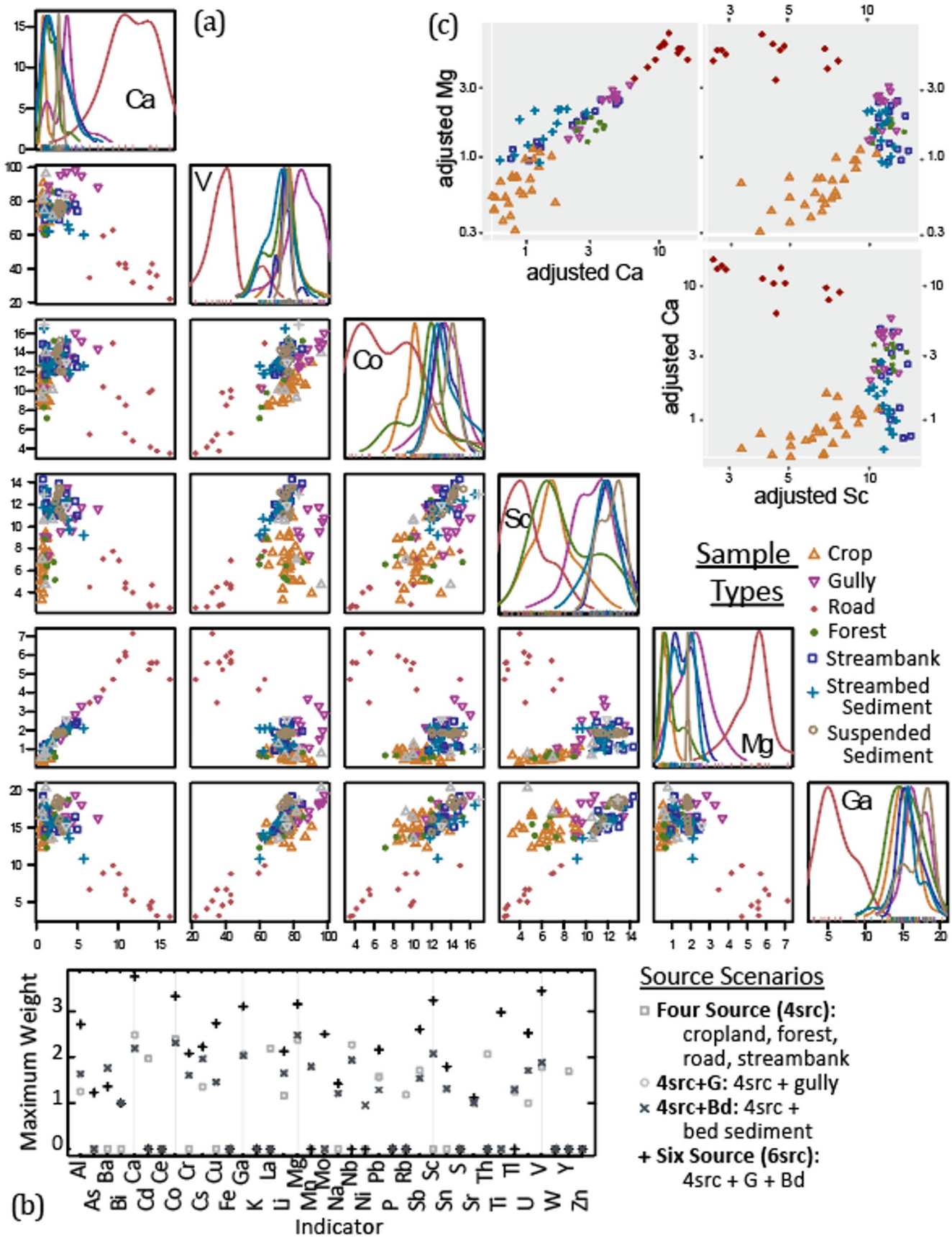
(a)
Ca
V
Co
Sc
Mg
Ga
(c)
adjusted Mg
adjusted Ca
adjusted Ca
adjusted Sc
Sample Types
Crop
Gully
Road
Forest
Streambank
Streambed Sediment
Suspended Sediment
Maximum Weight
Indicator
Al As Ba Bi Ca Cd Ce Co Cr Cs Cu Fe Ga K La Li Mg Mn Mo Na Nb Ni Pb P Rb Sb Sc Sn S Sr Th Ti Tl U V W Y Zn
(b)
Source Scenarios
Four Source (4src): cropland, forest, road, streambank
4src+G: 4src + gully
4src+Bd: 4src + bed sediment
Six Source (6src): 4src + G + Bd

Fig. 4 (**a**) Unadjusted concentration distribution and bi-plots of the six geochemical indicators most commonly used in the forward stepwise linear discriminant function analysis (LDA) for sediment fingerprinting; six outliers are shown in gray. These same geochemical indicators were also those with the largest (**b**) weight in the LDA. Note that seven indicators were not used for any samples, as indicated by a weight of zero for all scenarios. (**c**) Examples of the three adjusted-concentration bi-plots for indicators used by the LDA for the June 2022 sample collected with passive sampler. These particle size (d_{50}) and organic-carbon (OrgC) adjusted concentrations were plotted on a log axis to enable visual distinction of the non-road sources. Concentrations of Ca and Mg are in percent; concentrations of other elements are in ppm (more details in SI3)

indicators varied by as much as 29% (SI2). Differences in source verification for June and AugSep Passive replicates ranged from 0 to 13%, with all but one < 4% (Fig. 5).

The source-verification analysis indicated source confusion for at least one sample from each source type (Fig. 5). Cropland and road samples from each basin (East River, Apple Creek, Plum Creek) were consistently identified in all source scenarios although partial confusion occurred for some samples. Two forest samples were mis-identified depending on the scenario. Individual samples with the highest apportionment in the correct source type using the 4src scenario were similarly identified when gully (4src + G) and bed (4src + Bd) were introduced. Streambank samples showed confusion with every source but road, and some were mis-identified as either forest, gully, or bed depending on the source scenario (Fig. 5, SI2). Source verification of bed-sediment was similar to forest and gully (Fig. 5), supporting inclusion as a source. If other sources were confused as streambank for 4src, bed became the most likely source to cause confusion.

3.4 Suspended-Sediment apportionment using source scenarios

Across source scenarios and SS targets, in-channel sources (streambank + bed) were the largest contributor using either mean apportionments (Table 1; Fig. 6) or the range from the MC1000 iterations (SI4). Inclusion of bed sediment (6src and 4src + Bd) commonly resulted in decreased streambank apportionment (Fig. 6). Inclusion of the gully source affected apportionment of streambank as well as cropland and forest. Mean streambank apportionment was 0–100% among all source scenarios and targets, with apportionment in May and June lower for all scenarios (Fig. 6). June streambank apportionment was 48–67% with 4src and 4src + G. With the 6src scenario, 0% streambank was apportioned for May, with partial replacement by bed in June, as well. Among all months, mean bed apportionment was 0–56% with 6src and 0–22% with 4src + Bd. For context, passive-sampler replicates from June and AugSep had differences in streambank-apportionment (apportioned as other sources) of 25% & 1%, respectively, with 4src, 7% & 5% with 4src + G, 90% & 9% with 4src + Bed, and 42% and < 1% with 6src. While there were differences in streambank versus bed apportionment for concurrent replicates (Passive and AUTO), combined in-channel apportionment did not significantly differ for any of the source scenarios (p-value ≥ 0.4365).

Among all source scenarios, median cropland apportionment was 0%, with a mean of ≤ 3%, and a maximum of 13–33% (4src and 4src + Bd, respectively). Cropland was identified in June, AugSep, and OctNov samples (Fig. 5), corresponding to precipitation events (Fig. 2a). With inclusion of the gully source, mean cropland apportionment was 0–16% in the AUTO samples, but 0% for Passive samples. Mean gully apportionment was 1–32% using 6src and 4–32% using 4src + G. Mean forest apportionment was 0–36% when gully was a potential source and 0–84% if gully was omitted. Forest apportionment was higher during May and June (Fig. 5). Mean road apportionment was ≤ 4% among source scenarios and < 1% across all samples.

Although SS was infrequently apportioned as cropland- or road-sourced and only in small amounts, among MC1000 iterations these apportionments were generally more consistent (shorter range and wider violin-plot distribution in Fig. 6 and SI4) relative to other sources. The range of apportionments for individual samples reflects the source confusion from source verification, with infrequent iterations substituting forest and gully, or more commonly, streambank and bed sediment, depending on the sample (Fig. 5). Apportionment of forest, gully, streambank, and bed had more variability among iterations, with in-channel sources commonly substituting for each other among iterations (Fig. 6c-f).

4 Discussion

4.1 Comparison of suspended sediment collected with autosampler and passive sampler

There were no statistically significant differences between time- and flow-integrated samples collected with in-situ Passive sampler and AUTOsampler methods when compared as a function of physical (d_{50}) and chemical (OrgC and sed-P) characteristics. Both Passive and AUTO summer and fall samples were non-conservative in terms of sed-P but not samples from May and June, when sed-P was relatively lower and closer to source concentrations. Higher sed-P during low-streamflow conditions indicates P spiraling between dissolved and sediment-bound forms during transport (Kreiling et al. 2019; Simpson et al. 2021) combined with a lack of event-driven erosion and addition of fresh sediment to the stream (Williamson et al. 2021). This

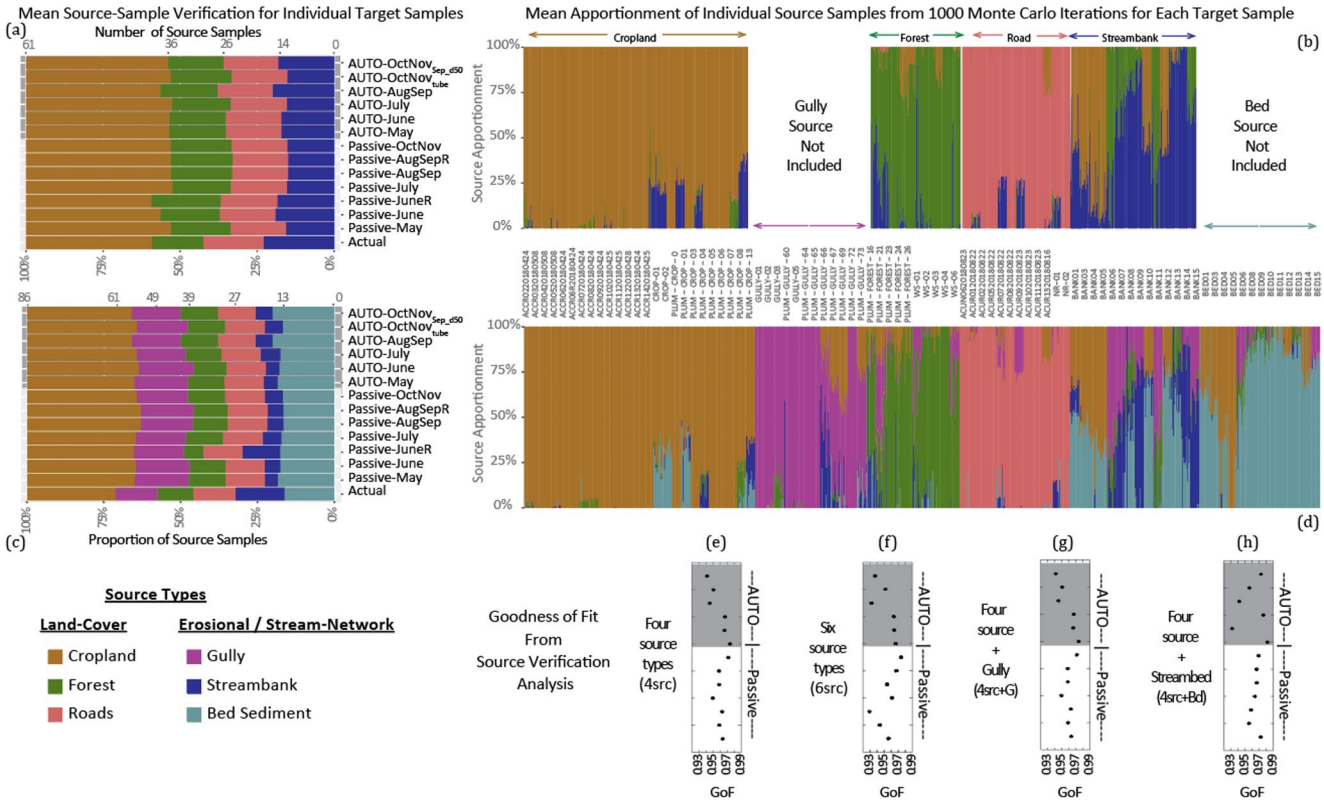

Fig. 5 Mean source verification and source confusion for individual source samples for each target sample. Examples for the four-source (4src; a,b) and the six-source (6src; c,d) are shown here; other scenarios are provided in SI2. Goodness of fit (GoF; e-h and Table 1), which evaluates accuracy of source-sample verification, did not significantly differ as a function of source scenario or collection method (automated-pump sampler [AUTO] or in-situ passive sampler [Passive])

sed-P enrichment is further indication that the SS population sampled was the same, regardless of method.

There were no patterns in GoF for source apportionment of SS as a function of collection method and the split between upland and in-channel sources did not significantly differ (Fig. 6). For example, using 4-src, streambank was consistently the largest proportional source regardless of collection method; using 6-src, in-channel sources were the maximum proportion for all but the June sample, with differences among the small upland-source proportions that reflect cropland-gully source confusion. *Combined with the overlap in d_{50}, OrgC, and sed-P, these results confirm that SS collected with Passive sampling represented the same population as that sampled by the AUTOsampler, like that used to estimate SS and particulate P loads, supporting our hypothesis*. Similar results were found in previous evaluations of mass and particle-size distribution of sediment collected in stream settings (Phillips et al. 2000; Goharrokhi et al. 2019) and sediment collection on cropland, although there were differences in P enrichment that were related to how much fine clay was captured (Panuska et al. 2011). However, results from this comparison of SS characteristics combined with the relative magnitude of source contributions, is foundational to communicating that sediment-fingerprinting results can be used to interpret sediment and sed-P loads, even in basins where SS is mostly silt and clay. For the range of streamflow conditions sampled, time-integrated samples collected by AUTO and Passive methods shared physical and chemical properties that resulted in similar source apportionment. Understanding SS source is critical in Great Lakes tributary basins where clay abundance directly influences both agricultural land management and stream-water P concentrations as a function of soil-water infiltration and movement, particle mobility, and the dynamic interaction of particulate & dissolved forms (Robertson et al. 2006).

4.2 Validation of geochemical indicators used and sources considered

Among all source apportionments, four to six source samples were eliminated as outliers for at least one geochemical indicator after each indicator was independently evaluated for normality. It is important to note that each source scenario was independent of the rest, however the same four crop samples were identified as outliers in all scenarios,

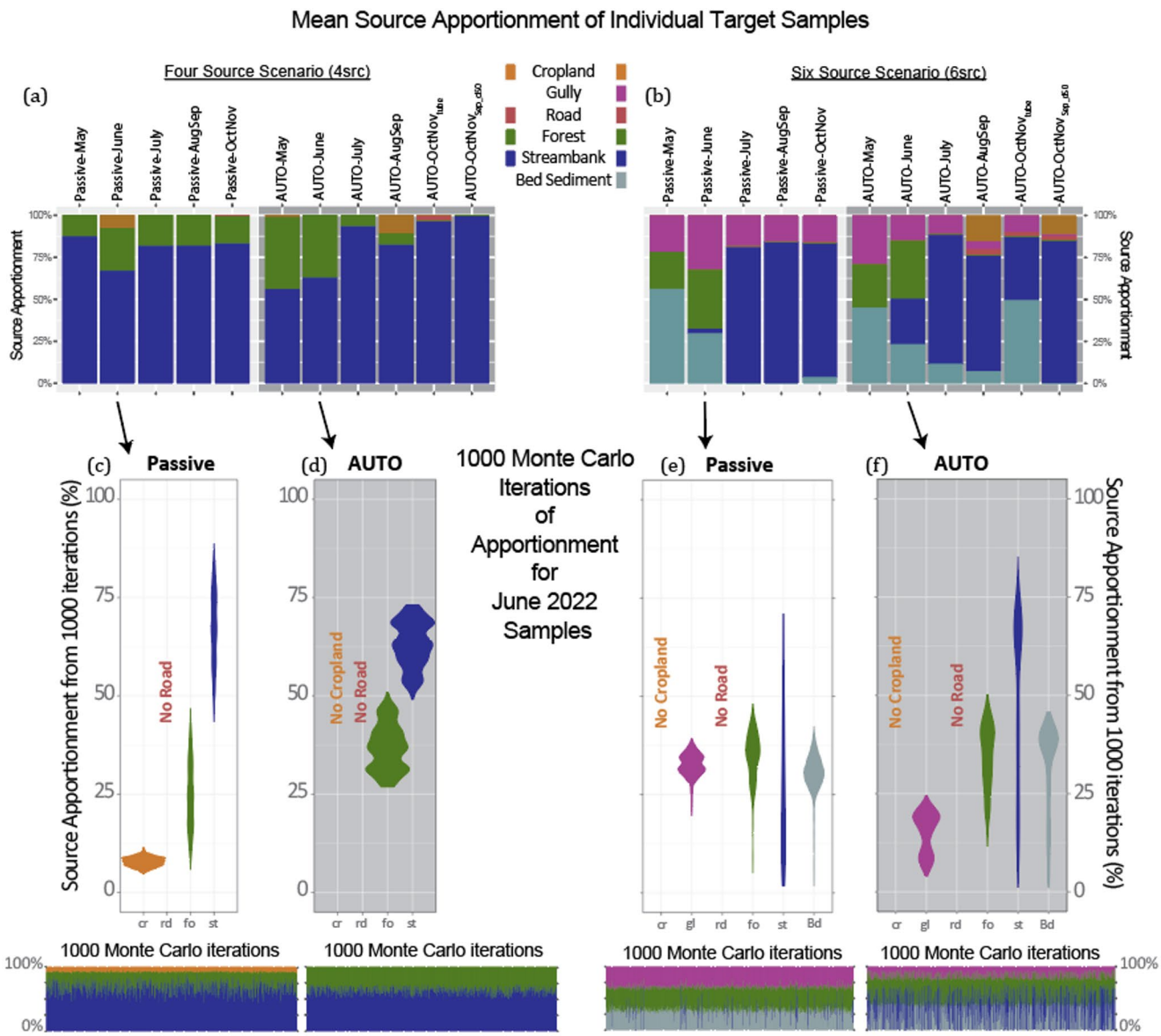

Fig. 6 Mean apportionments (**a**, **b**) of suspended-sediment samples using the four- (4src) and six- (6src) source scenarios with range of apportionments from 1000 Monte Carlo iterations for the June samples (**c**, **d**). The June sample is used as an example because it included the largest sampled event. Range of apportionments for all source scenarios, both sampling methods (AUTO, collected by automated-pump sampler; Passive, collected by in-situ passive sampler), and all sample periods are in SI4

even with addition of gully and bed source samples which had the potential to expand the overall concentration ranges of individual indicators. Omitting these outliers could be a source of uncertainty in source apportionment. One way of evaluating this would be to apportion the outliers using target-specific LDA, essentially including them in the source verification analysis, although this was beyond the scope of the current work. However, several factors indicate that omission of these four to six samples, relative to the 61–81 included, was unlikely to affect the apportionments. First, among the six most important geochemical indicators, outliers did not plot with other source groups (gray symbols in Fig. 4a), although they would have extended the maximum concentration for Co (Bed-2) and Ga (PL-CR02). Second, the coarse d_{50} of the East Basin cropland and gully source-sample outliers is further indication that these samples differed from the rest of the source group in terms of physical properties (Fig. 3a). Third, none of the geochemical indicators for which source samples were identified as outliers were among those six indicators most important to differentiating source groups (Fig. 4a). Outliers in a source dataset are not unusual. In a comparison of source-sample datasets (Williamson et al. 2023), outliers were highlighted for multiple source types, even though samples were evaluated

in the context of others from the same basin; this included outliers in the Apple Creek (1 road, 1 streambank) and Plum Creek (1 cropland, 1 streambank) datasets. The implication is that including these outliers would have minimally changed the source apportionments.

Geochemical indicators that differentiated sources in the East River Basin were a combination of those associated with plant material, nutrient application, exposed sub-soil, and by-products of vehicular use (which occurs on both cropland and roads). Of those geochemical indicators most commonly used for sediment fingerprinting East River SS (Fig. 4 and SI3), 4 of 6 (Ca, Co, Mg, V) are components of animal feed and are commonly two to three times enriched in the resultant dairy-cow manure (Sheppard and Sanipelli 2012) which is the most common cropland nutrient source in this basin. Besides manure, Ca and Mg are common in road brines used for snow and ice management (Casey et al. 2014), are plant macronutrients (Brady 1990), and are essential components of dolostone in the Niagara Escarpment (Luczaj 2016). Cobalt is a plant micronutrient (McBride 1994) that generally concentrates in clays along with other trace metals (Horowitz 1991), and it was the highest-weighted tracer in direct comparison of Apple Creek and Plum Creek samples (Williamson et al. 2023). Both Co and V complex with organic matter (McBride 1994), present as a result of conservation tillage and cover crops as well as forest leaf litter, and facilitate nitrogen fixation for both plants (Chatt et al. 1978) and algae (Pierzynski et al. 2000). Vanadium concentrations in soil generally increase from the surface- to sub-soil (Gustafsson 2019) and V correlates with Mg and Sc in the environment (Shaheen et al. 2019). Scandium is a rare earth element that occurs in many minerals in the Great Lakes landscape and is sometimes used as a reference element because it is not concentrated in geologic processes (Granmo et al. 2020; Wang et al. 2020). Gallium, which commonly substitutes for Al, Fe, and Zn in clay minerals (Foley et al. 2017; Granmo et al. 2020), was a geochemical indicator for sediment fingerprinting of all Plum Creek SS samples and combined with Sc for sediment fingerprinting of at least one SS sample in both Plum Creek and Apple Creek Basins (Fitzpatrick et al. 2019; Blount et al. 2022c). Both Sc and Ga concentrate in silt-sized material (Połedniok 2008).

Comparison of the unadjusted (Fig. 4a) and adjusted (Fig. 4c) concentrations for these most common indicators illustrates how the considered sources were differentiated. Association of Ca and Mg with multiple source types explains the linear relation between the two and why adjusted concentrations of these two geochemical indicators were discriminant factors for each SS target, as has been seen for other basins (Williamson et al. 2023). Overlap of streambank with forest, gully, and bed-sediment sources is evident in the biplots (Fig. 4a, c). Land-use / landform-based sediment fingerprinting could benefit from research on what governs the streambank geochemical fingerprint, which may be the absence of a land-use signature combined with stream-water chemistry and the geochemical composition of parent material as altered by soil development.

Sediment-fingerprinting GoF, a function of the source verification, was ≥ 0.93. However, source verification (Fig. 5) showed that some source types were better discriminated than others. As determined in other agricultural-basin studies (Williamson et al. 2014, 2020), cropland and road samples were most accurately identified in source verification, with the least source confusion, regardless of source scenario. This is further emphasized when looking at source verification of individual samples. The implication is that apportionment of SS as cropland- or road-sourced is based on well-defined sediment fingerprints (Fig. 4a, c). Forest, gully, and bed samples were also identified during source verification for most targets (SI2), but there was more source confusion when connectivity (4src + G) and storage (4src + Bd) sources were included (Figs. 4c and 5). The source verification of bed sediment indicates a distinct sediment fingerprint using the provided geochemical indicators. In contrast and regardless of source scenario, source verification of streambank samples indicated more source confusion than for other sources. These source verification results are consistent with other sediment-fingerprinting studies, including from Apple Creek and Plum Creek (Fitzpatrick et al. 2019; Blount et al. 2022c; Williamson et al. 2023).

4.3 Source-scenarios expand how sediment fingerprinting indicates connectivity between land use and the channel network to inform basin-management strategies

Sediment-fingerprinting studies are most useful when *a priori* identified sources can be explained in terms of process and seasonality. Field-based stream-corridor sediment budgets of Apple and Plum Creeks had documented both streambank erosion and accumulation of bed sediment in this post-glacial landscape (Fitzpatrick et al. 2019; Blount et al. 2022c). For Plum Creek, where lower order streams converge to a primary, entrenched valley (Fig. 1), streambank erosion and gully development were each identified in SS by sediment fingerprinting, along with infrequent and small (< 10%) forest (woodland) and cropland contributions. However, for Apple Creek, where the stream network is more dendritic (Fig. 1) and the basin has less relief, sediment fingerprinting apportioned cropland as the largest SS contributor for 8 of 12 months with streambank second. Ultimately, this indicates that the East River Basin is more

similar to Plum Creek Basin than to Apple Creek Basin, due to a combination of stream-network development and proximity to the Niagara Escarpment (Fig. 1). This finding aligns with previous work in Great Lakes basins that highlights that management may benefit from considering basin geomorphology (Williamson et al. 2024, 2025).

Recent cropland monitoring in the East River Basin documented erosion and transport of sediment from agricultural fields (Hanrahan et al. 2025) and water-quality modeling continues to highlight better management of agricultural lands as key to controlling nHABs (Merriman et al. 2019; Robertson et al. 2023). A complicating factor of integrating models with monitoring is that SS derived from fields can take multiple events to travel from source to basin outlet (Verhoff et al. 1979; Rose and Karwan 2021) creating a lag between mobilization from upland sources and delivery to the basin outlet and downstream water body (Haygarth et al. 2012; Kreiling et al. 2018). Understanding how much of the SS load is remobilized bed sediment from previous streamflow events addresses part of this knowledge gap.

We developed source scenarios to investigate this knowledge gap and to further explore the relative upland and in-channel contributions to SS in the East River. It is important to note that GoF for SS apportionment does not evaluate whether the correct sources were considered. The benefit of these scenarios was that each SS source could be evaluated as a function of how it was apportioned and what changed when connectivity (gullies) and storage (bed sediment) sources were omitted (Fig. 7).

Sediment fingerprinting started with the 4src scenario; this fundamental suite of land-use based sources with eroded streambank has proven useful in other regions (Boudreault et al. 2019; Xu et al. 2022) as well as Apple Creek and Plum Creek Basins. Inclusion of gullies (4src + G and 6src) addressed potential contributions from concentrated runoff that erodes vertically into the soil (USDA-NRCS 2002), providing a direct connection between uplands and the stream network. When not included as a source, gully-derived sediment would likely be apportioned as a combination of the land-use into which it incised, whether cropland

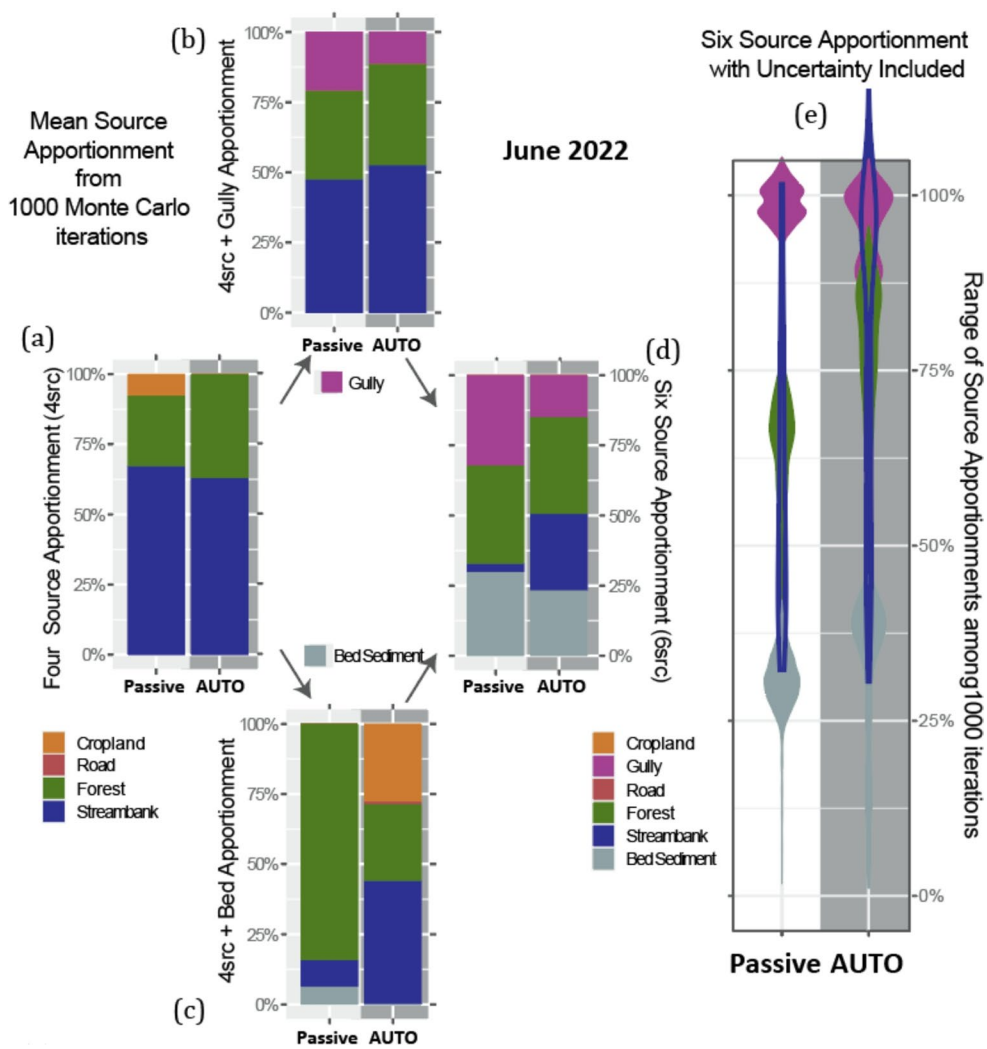

Fig. 7 Sediment fingerprinting of the June 2022 suspended-sediment replicates (Passive, collected by in-situ passive sampling; AUTO, collected by automated-pump sampler) as an example of how source scenarios can be compared to help inform management strategies. This benefits from considering both the mean apportionments (**a**-**d**) as well as (**e**) the uncertainty analysis enabled by the 1000 Monte Carlo iterations, stacked using the median of each apportionment. Source uncertainty between the concurrent replicates indicates that gullies connect multiple land uses and that in-channel sources can be the most difficult to differentiate

(USDA-NRCS 2002; Capra 2013), roads and development, or forested slopes; or as streambank-sourced, with which these features are sometimes sampled (Collins et al. 2010; Sherriff et al. 2018). Independent inclusion of East River bed sediment (4src + Bd and 6src) characterized the proportion of SS remobilized from in-channel storage as opposed to recent erosion of other sources, although it is unclear how long material might have been stored in the channel before remobilization. The fact that bed sediment is commonly evaluated as a sediment fingerprinting target (Williamson et al. 2014; Gellis et al. 2017; Cashman et al. 2018) aligns with the expectation that it is a mix of other sources. Consequently, when bed sediment was omitted as a source, contributions from the original sources would be expected to be higher.

Each source scenario indicated that upland contributions were a larger proportion in May and June, when event response was larger and while both cropland and forest canopies were still developing. This is consistent with results from Plum Creek (Fitzpatrick et al. 2019), although upland contributions in Apple Creek were the largest proportional contributors for most of the year (Blount et al. 2022c). Our samples do not include high, early spring streamflow periods, which have indicated higher in-channel sourced apportionments in other Great Lakes Basins (Williamson et al. 2020), including Plum Creek (Fitzpatrick et al. 2019). Using the 4src scenario, most upland-sourced sediment was apportioned as forest, with little apportionment of cropland (Fig. 6). Road apportionment was < 1%, however the distinct sediment fingerprint for this source indicates that its inclusion can improve the ability to discern other sources.

Inclusion of the gully source decreased cropland, forest, and streambank apportionments (Figs. 6 and 7), highlighting that gully-sourced material was present during all months sampled. This shift in sediment-fingerprint apportionment between 4src and 4src + G likely reflects that runoff concentrates in gullies as it moves from cropland, through forest, to the channel and underscores that water movement from uplands to the stream is a connection that could be considered for prioritization in addition to the cropland surface, where most conservation management is focused (Jurjonas et al. 2022). Conservation tillage and cover crops influence infiltration and soil-water storage (Moriasi et al. 2020). However, the general absence of cropland when the gully source is considered indicates a link to how water leaves the field. Tile drainage (Webber and Williamson 2021; Williamson et al. 2025) and road-runoff management (Fitzpatrick et al. 2019; Blount et al. 2022c, 2023) are designed to move excess water to streams, affecting stream response (Baker et al. 2004) and the resultant energy to erode slopes and streambanks (Zhou and Margenot 2025). Gully contributions continued to be differentiable with inclusion of the bed-sediment source, indicating that erosion in these connecting features is distinct from that in the rest of the stream network. This finding highlights the potential benefit of better understanding gully formation and distribution, how it might relate land and water management, and the seasonal connectivity of these features. Further investigation could use fallout radionuclides to independently differentiate these sources and estimate the recency of sediment mobilization relative to what is transported as SS (Karwan et al. 2018).

With the exception of June, most SS was from an in-channel source (streambank + bed; Fig. 5, SI4). Inclusion of the bed-sediment source most commonly decreased streambank, although total, in-channel source apportionment was similar (Figs. 5, 6 and 7). This indicates streambank as a major component of accumulated bed sediment, as was common in the Apple Creek and Plum Creek Basins (Fitzpatrick et al. 2019; Blount et al. 2022c). Applying June 4src and 4src + G apportionments, the month with the lowest in-channel apportionments and the highest monthly sediment load for the East River (Fig. 2), indicates that at least 3000 Mg yr^{-1} of sediment is derived from streambank; this approximates the reduction goal of 50%.

With the exception of the OctNov AUTO sample, for which d_{50} had to be estimated, bed-sediment apportionment was lower after June, indicating that during relatively high streamflow, sediment was resuspended from the channel bottom (Figs. 5 and 7a, c and d). As much as half of SS was remobilized bed sediment during May and June. However, during lower streamflow, there was less energy to remobilize bed sediment. Streambank erosion was the larger in-channel contributor as the seasons progressed and the smaller SS load (Fig. 2b) enabled discernment of sediment from upland sources. A possible explanation is that during the later months of sustained low streamflow, easily resuspended, finer-grained, bed sediment was exhausted while eroding streambanks continued to supply sediment, albeit in small amounts. Field surveys in Apple Creek and Plum Creek estimated in-channel bed-sediment storage at one to three times the annual SS load (Fitzpatrick et al. 2019; Blount et al. 2022c). Understanding East River bed-sediment sources and amount could provide additional information on how to manage sediment and sed-P delivery to the Fox River and Green Bay.

4.4 What was learned from considering source-scenario effect on source verification and target apportionment

Sediment-fingerprinting apportionments are commonly presented as mean values of multiple iterations (Collins et al. 2010). These representative numbers can be useful in communication with stakeholders. Unlike mean apportionments,

median apportionments are not guaranteed to sum to 100%. However, the established methods encapsulated in SedSAT (Gorman Sanisaca et al. 2017) enabled us to evaluate both the mean apportionments as well as the distribution among MC1000 iterations for each of the source scenarios for both source and target samples. The uncertainty illustrated by source verification (Fig. 5) and the consideration of all iterations for each SS apportionment (Figs. 6 and 7) are useful both in the range of sources identified as well as those that are not.

For example, in the month with the highest streamflow and most events (June; Fig. 2), cropland-sourced (7%) sediment was only identified in one of the concurrent replicates using the 4src scenario and no cropland-sourced sediment was apportioned using the 6src scenario (Fig. 6). There was little variability among MC1000 iterations for this cropland apportionment, and all iterations for both of the concurrent replicates identified streambank as the largest proportional contributor followed by forest. This difference in apportionment of the June concurrent replicates from the two sampling methods was within the 25% difference between replicates from passive-samplers using 4src; September passive-sampler replicates differed less. When gully was a considered source, ≤ 15% of sediment was apportioned as cropland among all SS samples and only after July, when streamflow and suspended sediment concentrations were lower. However, although the largest proportional source remained streambank, there was interaction among forest, gully, and cropland apportionments (Fig. 6 and SI4). This interaction reflects the overlap in the sediment fingerprint from these sources (Fig. 4) and confusion in source-sample identification (Fig. 5), as well as how they are connected in the landscape.

Sources of uncertainty considered here, including how SS is collected, which sources to consider, source confusion, and the distribution of apportionments provided by the MC1000 iterations, provide an alternative method of discussing sediment-fingerprinting results that may be helpful when findings seem to contradict other information. Management resources and basin-wide management plans commonly focus solely on cropland (Jurjonas et al. 2022; The Cadmus Group 2012). A potential implication of small and inconsistent apportionments is that a relatively small amount of cropland-sourced sediment is always present, but events and higher streamflow result in more gully and streambank erosion as well as re-suspension of bed sediment, diluting the cropland contribution and making it more difficult to detect. These findings reflect a WY2017 study in Plum Creek (when mean annual streamflow at the East River streamgage was 1.17 m^3 s^{-1}), where cropland was only differentiated at an upstream sampling site, but not closer to the confluence with the Fox River (Fitzpatrick et al. 2019). In contrast, in WY2018 in Apple Creek (0.97 m^3 s^{-1} at East River), apportionment of cropland at the downstream site exceeded that upstream (Blount et al. 2022c). This East River study (WY2022; 0.6 m^3 s^{-1}) occurred during a relatively dry year. For context, the maximum mean annual streamflow for the period of record was 1.9 m^3 s^{-1} (WY2019). During a wetter year, the cropland contribution may be even more difficult to identify because connectivity and in-channel sources could contribute even more of the SS load. Repeating the sampling described here during a wetter year or separately sampling event- and baseflow conditions could provide new information.

Further information could be provided by comparing source apportionment for co-incident and contemporaneously sampled suspended and bed sediment. What accumulates as bed sediment is a mix of upland and streambank from previous events (Cashman et al. 2018; Fitzpatrick et al. 2019; Blount et al. 2022c); this mixture results in a distinct sediment fingerprint for bed. This highlights that the accumulation of SS as bed sediment and remobilization of this stored material integrates a continuum of basin conditions, including different weather and streamflow patterns (Williamson et al. 2024), accumulation of sediment previously stored upstream (Gellis and Noe 2013), and a history of upland and stream-network management (Williamson et al. 2014). Consequently, continuing to expand our understanding of SS sources, landscape connectivity, and the lag between sediment mobilization from upland & in-channel sources and delivery to downstream rivers, lakes, and oceans can improve our ability to manage and protect these landscapes.

5 Conclusions

- Comparison of samples collected using Passive sampling, typically used for sediment-fingerprinting of SS, and an AUTOsampler, typically used for sampling SS and TP loads, showed no statistically significant differences in d_{50}, OrgC, or sed-P. Sample similarity extended to identification of the largest sediment sources, justifying the integration of sediment fingerprinting and loads results and how this is communicated to resource managers.
- Apportionment of land-use-sourced SS decreased when gully was included in source scenarios. This indicates that in relatively steep, clay-rich watersheds, these erosional features connect uplands and the channel network as a function of concentrated runoff. This highlights the potential of considering links between landscape management and what happens to water after it leaves the boundaries of a specific land-use.

- In-channel sources (streambank + bed sediment) supplied the largest proportion of SS, with larger bed apportionment during spring streamflow and more streambank apportionment in summer and fall. The consistent presence of these in-channel sources and proximity to the stream, combined with gullies which connect to the larger network, explain how the cropland signature for SS may be diluted between cropland fields and receiving water bodies.
- Considering different source scenarios, resultant source confusion, and how source apportionment varied provided new information on how sediment and water may travel from the uplands to the stream network. Ultimately, this shows the value of considering the hydrologic connectivity between upland land use, including cropland, and the stream network, where basins are sampled for monitoring.

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Author contributions Williamson, Fitzpatrick, Kreiling, and Komiskey designed this study and companion efforts in cooperation with the Great Lakes Restoration Initiative. Blount, Mavis, and Broerman did a majority of the field work. Hoefling and Pace managed laboratory analyses. Williamson did the structured analysis and most of the first draft, with input from co-authors and notable contributions from Broerman, Fitzpatrick, and Kreiling. Williamson, Fitzpatrick, and Kreiling led the revisions.

Data availability Data are available from U.S. Geological Survey's ScienceBase as Pace et al. (2025a-d).

Declarations

Competing interest The authors declare no competing interests.

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